

Nicholas Campbell

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Education

- MEng (Cum Laude) Stellenbosch University**, Mechatronic Engineering 2024 – 2025
- Thesis: An Architecture for a Federated Digital Twin System for an Automotive Enterprise (80%)
 - Conducted in close collaboration with BMW IT Hub South Africa
 - Presented conference paper at SOHOMA 2025, Aix-en-Provence, France (Oct 2025)
- BEng Stellenbosch University**, Mechatronic Engineering 2020 – 2023
- Best Final Year Biomedical Project — Institute of Biomedical Engineering (Wearable Cardiopulmonary Exercise System, 90%), Dec 2023

Experience

- SpatialEdge**, Junior Data Engineer 2026 – present
- Migrated legacy data infrastructure to a cloud-native architecture on GCP, ingesting on-premises relational databases, SaaS platforms (Salesforce, Jira), ERP systems, and internal case management systems into BigQuery
 - Architected and deployed an automated Salesforce → BigQuery ingestion pipeline using Python and BigQuery Data Transfer Service; built a CLI utility that auto-generates YAML configs from the Salesforce API, reducing new-asset onboarding from hours to minutes
 - Engineered BigQuery optimisations — transforming standard views into sharded tables and using table cloning to provide high-performance writable environments without additional storage costs
 - Automated GitLab CI/CD pipelines to streamline deployment cycles and synchronise configurations with cloud infrastructure, incorporating dual-stage validation and POPIA compliance tagging
 - Led production deployments and participated fully in Agile sprint ceremonies
- BMW IT Hub South Africa**, Graduate Researcher 2024 – 2025
- Collaborated with multiple stakeholders to derive business requirements, translating them into technical requirements for a system.
 - Designed software architectures integrating heterogeneous digital twin platforms across the vehicle lifecycle, applying cloud services and knowledge graphs for DevOps integration and lifecycle analytics.
 - Participated in the BMW Automotive IT Colloquium (Jun 2025) on AI, Digital Twins, and Automation; contributed to a hackathon building a VS Code extension that integrates an LLM with the GitHub API to enhance developer efficiency across the DevOps lifecycle
- Stellenbosch University**, Head Teaching Assistant — Mechatronics 424 2025 – 2025
- Oversaw execution of Mechatronics 424 group projects; coordinated demonstrators, managing tasks, timelines, and team communication
- Zutari**, Engineering Intern 2022 – 2023
- Rotated through Industrial Automation and Water teams
 - Developed a GeoFencing application in Ignition Perspective using phone-based location data to demonstrate safety and control strategies

Projects

Personal Finance Tracker

2026

- Full-stack personal finance dashboard built with Next.js (TypeScript), Supabase (PostgreSQL), and deployed on Vercel; ingests bank transactions nightly via a Gmail Apps Script integration
- Engineered a nightly ETL pipeline (raw → prepared) with multi-currency FX conversion, a three-pass categorisation engine (rules → LLM → manual overrides), and Vercel Cron Jobs for scheduling
- Features net worth tracking, budget management, recurring transaction detection, and a GitHub-style spend heatmap across 8 dashboard pages

Wearable Cardiopulmonary Exercise System

2023

- Designed, built, and tested a low-cost mobile CPET system with wireless real-time data streaming up to 90 m; strong correlation with reference system
- Awarded Best Final Year Biomedical Project by the Institute of Biomedical Engineering (90%)

GeoFencing Application

2023

- Built a concept demonstrator in Ignition Perspective during a vacation-work period at Zutari, using phone-based GPS data to implement safety and control strategies for the Industrial Automation team

Analog Audio Amplifier

2023

- Designed, built, and tested a class AB 2-channel audio amplifier outputting 8W as part of Electronics 315 (undergraduate mechatronics)

Skills

Languages: Python, SQL, JavaScript, C

Databases: BigQuery, PostgreSQL, MongoDB, Firebase, Supabase

Cloud & Infrastructure: Google Cloud Platform, Apache Airflow, GitLab CI/CD, GitHub Actions, Docker; some experience with AWS & Azure

Data Engineering: ETL pipeline development, data warehouse optimisation, CI/CD automation

Mechatronics: Embedded systems, CAD, control systems, signal processing

Domain Knowledge: Systems integration, software architectures for data aggregation, digital twins, knowledge graphs

Professional: Agile methodology, sprint planning, cross-functional collaboration, technical documentation, production deployment management

Awards

Best Final Year Biomedical Project: Institute of Biomedical Engineering, Dec 2023 — Wearable Cardiopulmonary Exercise System (90%)

Data School Hackathon — 2nd Place: Standard Bank & Mobalyz, Aug 2023 — real-time driver risk-scoring system

Golden Key Society: Invited 2021, 2022, 2023